



ARTÍCULO DE INVESTIGACIÓN

Hypersurfaces with Prescribed Boundary and Small Steklov Eigenvalues

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Resumen

Dada una hipersuperficie suave y compacta M con frontera $\Sigma = \partial M$, demostraremos la existencia de una sucesión M_j de hipersuperficies con la misma frontera que M , tal que cada valor propio Steklov $\sigma_k(M_j)$ tiende a cero cuando j tiende al infinito. Las hipersuperficies M_j se obtienen a partir de M mediante una perturbación local cerca de un punto de su frontera. Sus volúmenes y diámetros son arbitrariamente cercanos a los de M , mientras que las curvaturas principales de la frontera permanecen inalteradas.

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1. Introducción

Let M be an n -dimensional smooth compact Riemannian manifold with boundary $\Sigma = \partial M$. The Steklov eigenvalue problem on M consists in finding all numbers $\sigma \in \mathbb{R}$ for which there exists a nonzero function

$u \in C^\infty(M)$, which solves

$$\begin{cases} \Delta u = 0 & \text{in } M, \\ \partial_\nu u = \sigma u & \text{on } \Sigma. \end{cases}$$

Here, Δ is the Laplacian induced from the Riemannian metric g on M , and ∂_ν is the outward pointing normal derivative along the boundary Σ . The Steklov eigenvalues form an unbounded increasing sequence $0 = \sigma_0 \leq \sigma_1 \leq \sigma_2 \leq \dots \rightarrow \infty$, each of which is repeated according to its multiplicity. Note that if M is connected, then $\sigma_1 > 0$. See [8, 12] for background on this problem.

One of our main interests in recent years has been to understand the particular role that the boundary Σ plays with respect to Steklov eigenvalues. Some papers studying this question are [4, 14, 6, 2, 16, 11, 9, 5, 15]. In particular, we have considered the effect of various geometric constraints on individual eigenvalues σ_k . One particularly interesting question is to prescribe a Riemannian metric g_Σ on the boundary Σ and to investigate lower and upper bounds for the eigenvalue σ_k among all Riemannian metrics g that coincide with g_Σ on the boundary. Given any Riemannian metric g on M such that $g = g_\Sigma$ on Σ , it is proved in [6] that one can make any eigenvalue σ_k arbitrarily small by modifying the Riemannian metric g in an arbitrarily small neighborhood $V \subset M$ of a point $p \in \partial M$. More precisely, for each $\epsilon > 0$ and each $k \in \mathbb{N}$, there exists a Riemannian metric $\tilde{g} = \tilde{g}_{\epsilon,k}$ on M that coincides with g_Σ on Σ and also with g outside the neighborhood V , such that $\sigma_k(M, \tilde{g}) < \epsilon$. In [5] a more restrictive constraint was imposed by requiring the manifold M to be a submanifold of $\mathbb{R}^m$ with prescribed boundary $\Sigma = \partial M \subset \mathbb{R}^m$. In this context an upper bound for σ_k was given in terms of Σ and of the volume of M . The authors were unable at the time to give a lower bound and they raised the question of whether one exists, or if instead, arbitrarily small eigenvalues are possible. The goal of this paper is to answer that question.

Teorema 1.1. *Let $M \subset \mathbb{R}^{n+1}$ be a smooth n -dimensional compact hypersurface with nonempty boundary $\Sigma = \partial M$. For each $p \in \Sigma$, there exists a sequence of hypersurfaces $M_j \subset \mathbb{R}^{n+1}$, $j \in \mathbb{N}$, with boundary $\partial M_j = \Sigma$ and with the hypersurface M_j coinciding with M outside of a ball $B(p, \frac{1}{j})$, such that*

$$\lim_{j \rightarrow \infty} \sigma_k(M_j) = 0 \quad \text{for each } k \in \mathbb{N}. \quad (1.1)$$

The principal curvatures of $\Sigma \subset M_j$ are independent of j . Moreover, the volume and diameter of M_j converge to those of M as $j \rightarrow \infty$.

In order for (1.1) to hold for each $k \in \mathbb{N}$, it is necessary that the perturbed hypersurfaces M_j differ from M arbitrarily close to the boundary Σ as $j \rightarrow \infty$. Indeed, let b be the number of connected components of Σ and note that any hypersurface $\tilde{M}$ that coincides with M in a neighborhood Ω of Σ satisfies $\sigma_{b+1} \geq C > 0$, where C is given by a sloshing problem on $\Omega \cap M$; see [5] for details.

Observación 1.2. Teorema 1.1 holds in arbitrary positive codimension and ambient space. We decided to state it for hypersurfaces in $\mathbb{R}^{n+1}$ for the sake of notational simplicity. Note also that Teorema 1.1 certainly holds under weaker regularity assumptions.

Observación 1.3. By construction (see Section 3), each manifold M_j coincides with M in a neighborhood Ω_j of its prescribed boundary Σ . Hence, the Dirichlet-to-Neumann maps $D_j: C^\infty(\Sigma) \rightarrow C^\infty(\Sigma)$ associated with M_j all have the same full symbol [13]. The asymptotic behavior of $\sigma_k(M_j)$ as $k \rightarrow \infty$ is therefore independent of j . That is, for each $j_1, j_2 \in \mathbb{N}$ the following holds: $\sigma_k(M_{j_1}) - \sigma_k(M_{j_2}) = O(k^{-\infty})$; see [9, Lema 2.1]. In particular, the limits of $\sigma_k(M_j)$ as $j \rightarrow \infty$ and as $k \rightarrow \infty$ do not commute.

1.1. La Estrategia de la Demostración

For eigenvalues of the Laplace operator, it is well known that one can obtain arbitrarily small eigenvalues by constructing thin Cheeger dumbbells in the interior of the manifold; see [1, 3]. This strategy does not work for Steklov eigenvalues. For Steklov eigenvalues, it is possible to obtain arbitrarily small

eigenvalues by creating thin channels, but this involves deformation of the boundary or a perturbation of the Riemannian metric in the interior of M ; see [7, 6]. In order to prove Teorema 1.1, we have to use a more elaborate strategy.

Given a smooth function $\tilde{u}: \mathbb{R}^{n+1} \rightarrow \mathbb{R}$, consider the restriction $u = \tilde{u}|_M$. It is well known that if $\int_{\Sigma} u \, dA = 0$,

$$\sigma_1(M) \int_{\Sigma} u^2 \, dA \leq \int_M |\nabla u|^2 \, dV; \quad (1.2)$$

see [8] and Section 2 below. Here ∇u is the tangential gradient of u . It is the projection of the ambient gradient $\bar{\nabla} \tilde{u}$ on the tangent spaces of $M \subset \mathbb{R}^{n+1}$. The basic idea of our demostración is to fix a function $\tilde{u} \in C^\infty(\mathbb{R}^{n+1})$ and consider the vector field $\bar{\nabla} \tilde{u}$ in the ambient space $\mathbb{R}^{n+1}$. The hypersurface M is then deformed by creating “wrinkles” that tend to make the various tangent spaces $T_p M$, for $p \in \text{int } M$, perpendicular to $\bar{\nabla} \tilde{u}(p)$. This is achieved by “folding the surface like an accordion” in the direction perpendicular to $\bar{\nabla} \tilde{u}$. In the limit the right-hand-side of inequality (1.2) tends to zero. Let us illustrate this strategy with a simple example.

Ejemplo 1.4. Given a smooth function $f: \bar{\mathbb{D}} \rightarrow \mathbb{R}$ vanishing on the circle $S^1 = \partial \mathbb{D}$, consider the surface

$$S_f := \text{Graph of } f = \{(x, y, f(x, y)) : (x, y) \in \bar{\mathbb{D}}\}.$$

The boundary of S_f is the same for each f . We will use the function defined by $\tilde{u}(x, y, z) = x$ and its restriction $u = \tilde{u}|_{S_f}$ as a trial function in inequality (1.2). Because $\nabla \tilde{u} = (1, 0, 0)$, it follows from Lema 2.4 that the Dirichlet energy of $u := \tilde{u}|_{S_f}: S_f \rightarrow \mathbb{R}$ is given by

$$\int_{S_f} |\nabla u|^2 = \int_{\mathbb{D}} \frac{1 + f_y^2}{\sqrt{1 + f_x^2 + f_y^2}} \, dx dy.$$

For $n \in \mathbb{N}$, define $f = f_n: \bar{\mathbb{D}} \rightarrow \mathbb{R}$ by

$$f(x, y) = \sin(nx) \overbrace{(1 - x^2 - y^2)}^{\phi(x, y)}.$$

It follows from

$$f_x^2 = n^2 \left(\cos(nx) \phi + \frac{1}{n} \sin(nx) \phi_x \right)^2 \quad \text{and} \quad f_y^2 = \sin^2(nx) \phi_y^2$$

that

$$\lim_{n \rightarrow \infty} \int_{S_{f_n}} |\nabla u|^2 = 0.$$

Together with (1.2), this shows that $\lim_{n \rightarrow \infty} \sigma_1(S_{f_n}) = 0$.

La demostración del Teorema 1.1 is based on the above idea, but it is technically more involved, because we want to localize this argument to a small neighbourhood of a point p of the boundary. This is a significant gain compared to the above example, because it allows the construction of an arbitrary finite number of disjointly supported trial functions with small Dirichlet energy, leading to the collapse of each eigenvalue σ_k rather than just σ_1 . For the sake of readability and simplicity, the deformations that we use in the demostración del Teorema 1.1 are Lipschitz continuous but only piecewise smooth.

Plan del Artículo

In Section 2 we review the min-max characterization of Steklov eigenvalues, and we prove a lemma regarding the control of the Dirichlet energy under quasi-isometries. We then proceed to construct the perturbed hypersurfaces in Section 3. We use a quasi-isometric chart to a hypersurface with a flat boundary. The perturbed submanifold is then constructed by considering the graph of a locally supported oscillating function. Finally, in Section 4 an appropriate trial function is used to conclude the demostración del Teorema 1.1.

2. Notación y Consideraciones Preliminares

Let M be a smooth compact manifold with boundary Σ . The volume form on M is written dV , while the volume form on Σ is dA . We denote by $H^1(M)$ the standard Sobolev space of functions in $L^2(M, dV)$ with weak first derivative in $L^2(M, dV)$. The Steklov eigenvalues σ_k admits a variational characterization in terms of the *Steklov-Rayleigh quotient* of a function $0 \neq u \in H^1(M)$,

$$\mathcal{R}(u) = \frac{\int_M |\nabla u|^2 dV}{\int_\Sigma u^2 dA}.$$

The numerator $D(u) = \int_M |\nabla u|^2 dV$ is the *Dirichlet energy* of $u \in H^1(M)$. It is well known that

$$\sigma_k = \min_{\substack{S \subset H^1(M) \\ \dim S = k+1}} \max_{u \in S \setminus \{0\}} \mathcal{R}(u), \quad (2.1)$$

where the minimum is taken over all $(k+1)$ -dimensional linear subspaces $S \subset H^1(M)$.

2.1. Quasi-isometries and Dirichlet Energy

Let M and $\tilde{M}$ be two n -dimensional Riemannian manifolds with boundary. A diffeomorphism $\phi: M \rightarrow \tilde{M}$ is a *quasi-isometry* with constant $C \geq 1$ if for each $p \in M$ and each $0 \neq v \in T_p M$,

$$\frac{1}{C} \leq \frac{\|D_p \phi(v)\|^2}{\|v\|^2} \leq C.$$

Quasi-isometries provide a control of the Dirichlet energy of a function.

Lema 2.1. *Let $\phi: M \rightarrow \tilde{M}$ be a quasi-isometry with constant $C \geq 1$. Let $f \in H^1(\tilde{M})$; then*

$$\frac{1}{C^{\frac{n}{2}+1}} \leq \frac{\|\nabla(f \circ \phi)\|_{L^2(M)}^2}{\|\nabla f\|_{L^2(\tilde{M})}^2} \leq C^{\frac{n}{2}+1}.$$

Demostración. Let $\tilde{g}$ be the Riemannian metric of $\tilde{M}$ and let g be that of M . Let $\hat{g} = \phi^*(\tilde{g})$ be the pull-back of the metric $\tilde{g}$. Because ϕ is a quasi-isometry with constant C , the following holds for each $0 \neq v \in TM$,

$$\frac{1}{C} \leq \frac{g(v, v)}{\hat{g}(v, v)} \leq C.$$

It follows that

$$\frac{1}{C} g(\nabla_g f, \nabla_g f) \leq \hat{g}(\nabla_{\hat{g}} f, \nabla_{\hat{g}} f) \leq C g(\nabla_g f, \nabla_g f).$$

The corresponding volume forms satisfy

$$C^{-n/2} dV_g \leq dV_{\widehat{g}} \leq C^{n/2} dV_g.$$

This leads to

$$\begin{aligned} \|\nabla(f \circ \phi)\|_{L^2}^2 &= \int_M g(\nabla_g(f \circ \phi), \nabla_g(f \circ \phi)) dV_g \\ &\leq C^{n/2+1} \int_M \widehat{g}(\nabla_{\widehat{g}}(f \circ \phi), \nabla_{\widehat{g}}(f \circ \phi)) dV_{\widehat{g}} \\ &= C^{n/2+1} \int_{\widehat{M}} \widetilde{g}(\nabla_{\widetilde{g}} f, \nabla_{\widetilde{g}} f) dV_{\widetilde{g}}. \end{aligned}$$

The proof of the lower bound is identical, and accordingly omitted. $\square$

2.2. Quasi-isometric Charts

Recall that a subset $M \subset \mathbb{R}^{n+1}$ is a hypersurface with boundary if for each $p \in M$, there exist open sets $W, W' \subset \mathbb{R}^{n+1}$ with $p \in W$ and a diffeomorphism $\psi : W \rightarrow W'$ such that $\psi(M \cap W)$ is an open set in the half-space

$$H = \{x \in \mathbb{R}^{n+1} : x_{n+1} = 0, x_1 \geq 0\}.$$

The point $p \in M$ is on the boundary Σ of M if and only if ψ sends it to the boundary of the half-space H :

$$\psi(p) \in \partial H := \{x \in H : x_1 = 0\}.$$

This definition is coherent. It does not depend on the choice of the diffeomorphism ψ ; see [10, Chapter 1] for details. By further restricting ψ and scaling if necessary, we can assume that it is a quasi-isometry and that its image W' is a cylinder. This is summed up in the next lemma.

Lema 2.2. *For each $p \in \Sigma$, there exists a quasi-isometry*

$$\psi : W \longrightarrow W' = B_{\mathbb{R}^n}(0, 1) \times (-1, 1)$$

with $\psi(p) = 0$ and such that the image of $M \cap W$ is

$$U := \psi(M \cap W) = \{x \in H : |x| < 1\}.$$

Observación 2.3. We identify $U \subset \mathbb{R}^{n+1}$ with a subset of $\mathbb{R}^n$ so that we can write $x = (x_1, \dots, x_n) \in U$ instead of $x = (x_1, \dots, x_n, 0) \in U$.

In particular, the restriction of ψ to $\Sigma \cap W$ is also a quasi-isometry.

2.3. Dirichlet Energy on the Graph of a Function

Let $U \subset \mathbb{R}^n$ be a bounded open set and let $f : U \rightarrow \mathbb{R}$ be a bounded smooth function. Consider the graph

$$S_f = \{(x, f(x)) : x \in U\} \subset \mathbb{R}^{n+1}.$$

Given a function $u : U \rightarrow \mathbb{R}$, define $\widetilde{u} : U \times \mathbb{R} \rightarrow \mathbb{R}$ by $\widetilde{u}(x, x_{n+1}) = u(x)$ and define $u_f : S_f \rightarrow \mathbb{R}$ by

$$u_f(x, f(x)) = u(x) = \widetilde{u}|_{S_f}. \quad (2.2)$$

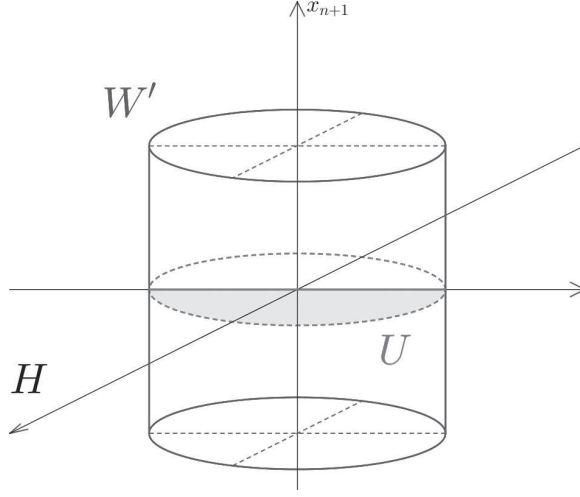


Figura 1. The domain U .

Lema 2.4. The Dirichlet energy of $u_f: S_f \rightarrow \mathbb{R}$ is

$$\int_{S_f} |\nabla u_f|^2 dV = \int_U \frac{|\nabla u|^2 + |\nabla u|^2 |\nabla f|^2 - \langle \nabla u, \nabla f \rangle^2}{\sqrt{1 + |\nabla f|^2}} dx, \quad (2.3)$$

where on the left-hand-side ∇ , dV , and the norm are taken on S_f , and on the right-hand-side $dx = dx^1 \cdots dx^n$ is the Lebesgue measure on U , while ∇ is the usual gradient on $\mathbb{R}^n$.

The Cauchy–Schwarz inequality gives $|\nabla u|^2 |\nabla f|^2 - \langle \nabla u, \nabla f \rangle^2 \geq 0$ with equality if and only if $\nabla u = c \nabla f$ for some constant c .

Demostración del Lema 2.4. To simplify notation, we will write $S = S_f$. For any point $p \in S$, the gradient $\nabla u_f \in T_p S$ is the projection of $\bar{\nabla} \tilde{u}$ on $T_p S$, that is,

$$\nabla u_f = \bar{\nabla} \tilde{u} - \langle \bar{\nabla} \tilde{u}, N \rangle N,$$

where N is a unit normal vector to $T_p S$. It follows from

$$\begin{aligned} \bar{\nabla} \tilde{u} &= \left(\frac{\partial u}{\partial x_1}, \dots, \frac{\partial u}{\partial x_n}, 0 \right), \\ N &= \frac{1}{\sqrt{1 + |\nabla f|^2}} \left(e_{n+1} - \sum_{i=1}^n \frac{\partial f}{\partial x_i} e_i \right). \end{aligned}$$

that

$$|\nabla u_f|^2 = \frac{|\nabla u|^2 + |\nabla u|^2 |\nabla f|^2 - \langle \nabla u, \nabla f \rangle^2}{1 + |\nabla f|^2}. \quad (2.4)$$

The volume element on S_f is given by

$$dV = \sqrt{1 + |\nabla f|^2}. \quad (2.5)$$

Together with identity (2.4), this completes the proof. $\square$

3. Perturbation of the Submanifold M

Given $p \in \Sigma$, let ψ be the quasi-isometric chart provided by Lema 2.2. In order to prove Teorema 1.1, we will deform the submanifold M in the neighborhood W of the point p by deforming the neighborhood $U \subset W'$ inside W' and pulling back to W using the quasi-isometry ψ . Consider a smooth function $f: U \rightarrow \mathbb{R}$ that is supported in the interior of U and which satisfies $|f(x)| < 1$ for each $x \in U$. This last condition implies that the graph of f ,

$$S_f = \{(x, f(x)) : x \in U\},$$

is contained in the cylinder W' . Hence it can be used to define a deformation of M as follows:

$$\tilde{M}_f := (M \setminus W) \cup \psi^{-1}(S_f). \quad (3.1)$$

Observación 3.1. It might be possible to prove Teorema 1.1 by performing a deformation of M directly in the ambient space $\mathbb{R}^{n+1}$, but it appears to be simpler to use quasi-isometric charts.

3.1. Deformation Function

We now construct specific functions f and u such that the Dirichlet energy of u_f , defined by (2.2), is small. Our method is based on Lema 2.4, which shows that if ∇u and ∇f are parallel the numerator of (2.3) is independent of ∇f , while the denominator behaves as $|\nabla f|$. Thus, we want f and u to have parallel gradients with $|\nabla f|$ big to get a small Dirichlet energy for u_f .

Consider numbers $\epsilon, \partial_1, \partial_2, \rho > 0$ that are sufficiently small and define $\partial := \partial_1 + \partial_2$. These constants will be adjusted later in equation (4.6). Let $q = (\partial, 0, \dots, 0) \in H$. Consider the following subsets of U :

$$\begin{aligned} A &:= \{(x_1, \dots, x_n) \mid x_1 \geq \partial, \|x - q\| \leq \epsilon\}, \\ B &:= \{(x_1, \dots, x_n) \mid \partial_1 \leq x_1 \leq \partial, \|\Pi x\| \leq \epsilon\}, \\ C &:= \{(x_1, \dots, x_n) \mid 0 \leq x_1 \leq \partial_1, \|\Pi x\| \leq \epsilon\}, \\ D &:= \{x \in U \setminus (A \cup B \cup C) \mid d(x, A \cup B \cup C) \leq \rho\} \end{aligned}$$

where $\Pi(x_1, x_2, \dots, x_n) = (x_2, \dots, x_n)$ is the projection on the boundary. Let $\Omega = A \cup B \cup C$. Vea la Figura 2, where the x_1 -axis is vertical.

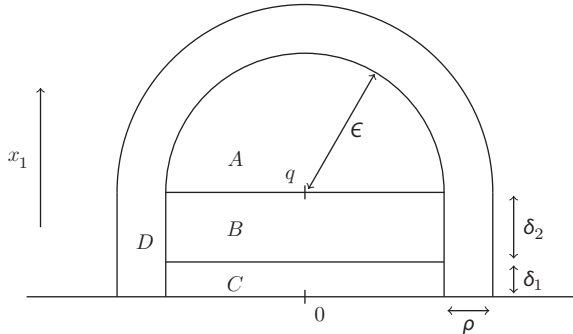


Figura 2. Perturbation region.

Define cutoff functions

$$\begin{aligned} \eta: [0, \infty) &\rightarrow \mathbb{R}, & \gamma: [0, \partial] &\rightarrow \mathbb{R}, \\ \eta(t) &= \max\{0, 1 - t\}, & \gamma(t) &= \begin{cases} 0 & \text{if } t \leq \partial_1, \\ t - \partial_1 & \text{if } \partial_1 \leq t \leq \partial. \end{cases} \end{aligned}$$

Given $\omega > 0$, define $f: U \rightarrow \mathbb{R}$ by

$$f(x) = \begin{cases} \eta\left(\frac{d(x, \Omega)}{\rho}\right) \gamma(x_1) F(\omega \|\Pi x\|) & \text{if } x_1 \leq \partial, \\ \eta\left(\frac{d(x, \Omega)}{\rho}\right) \partial_2 F(\omega \|x - q\|) & \text{if } x_1 \geq \partial. \end{cases} \quad (3.2)$$

4. Trial Function

The trial function u is supported on $\Omega \subset U$ and is defined by

$$u(x) = \begin{cases} 1 - \frac{\|\Pi x\|}{\epsilon} & \text{if } x \in B \cup C, \\ 1 - \frac{\|x - q\|}{\epsilon} & \text{if } x \in A, \\ 0 & \text{elsewhere.} \end{cases} \quad (4.1)$$

By construction, the function u_f defined by (2.2) belongs to $H^1(S_f)$, and we can estimate its Dirichlet energy. On A , the Dirichlet energy of u_f can be made small by taking ω big. Indeed, for almost all $x \in A$,

$$\nabla f = \pm \partial_2 \omega \frac{x - q}{\|x - q\|}, \quad (4.2)$$

$$\nabla u = -\frac{1}{\epsilon} \frac{x - q}{\|x - q\|}, \quad (4.3)$$

and using the fact that ∇f and ∇u are parallel, the Dirichlet energy is

$$\begin{aligned} \int_{S_f \cap A \times \mathbb{R}} |\nabla u_f|^2 dV &= \int_A \frac{|\nabla u|^2}{\sqrt{1 + |\nabla f|^2}} dx = \frac{1}{\epsilon^2} \frac{1}{\sqrt{1 + \partial_2^2 \omega^2}} \text{Vol } A \\ &= \frac{c_1 \epsilon^{n-2}}{\sqrt{1 + \partial_2^2 \omega^2}} \end{aligned}$$

where c_1 is some dimensional constant.

On B and C , ∇f and ∇u are not parallel, but it is possible to make the Dirichlet energy small by making the volume of B and C small. For B , we have for almost all $x \in B$:

$$\nabla f(x) = F(\omega \|\Pi x\|) e_1 \pm \gamma(x_1) \omega \frac{\Pi x}{\|\Pi x\|}, \quad (4.4)$$

$$\nabla u(x) = -\frac{1}{\epsilon} \frac{\Pi x}{\|\Pi x\|}, \quad (4.5)$$

and since e_1 and Πx are orthogonal,

$$|\nabla f(x)|^2 = F(\omega \|\Pi x\|)^2 + \gamma(x_1)^2 \omega^2.$$

Then the Dirichlet energy on B is

$$\begin{aligned}
 \int_{S_f \cap B \times \mathbb{R}} |\nabla u_f|^2 dV &= \int_B \frac{\frac{1}{\epsilon^2} + \frac{1}{\epsilon^2} F(\omega \|\Pi x\|)^2}{\sqrt{1 + F(\omega \|\Pi x\|)^2 + \gamma(x_1)^2 \omega^2}} dx \\
 &\leq \frac{1}{\epsilon^2} \int_B \frac{2}{\sqrt{1 + \gamma(x_1)^2 \omega^2}} dx \\
 &= c_2 \epsilon^{n-3} \int_0^{\partial_2} \frac{1}{\sqrt{1 + x_1^2 \omega^2}} dx_1 \\
 &= c_2 \epsilon^{n-3} \frac{\ln(\partial_2 \omega + \sqrt{1 + \partial_2^2})}{\omega},
 \end{aligned}$$

where c_2 is a constant that depends only on the dimension. And on C , since $f = 0$, the Dirichlet energy is simply

$$\int_{S_f \cap C \times \mathbb{R}} |\nabla u_f|^2 dV = \frac{1}{\epsilon^2} \text{Vol } C = c_3 \partial_1 \epsilon^{n-3},$$

where c_3 is a constant that depends only on the dimension. The denominator in the Steklov–Rayleigh quotient of u_f satisfies

$$\int_{B(0, \epsilon)} \left(1 - \frac{|x|}{\epsilon}\right)^2 dx \geq \frac{1}{4} \text{Vol} \left(B\left(0, \frac{\epsilon}{2}\right)\right) = c_4 \epsilon^{n-1},$$

for some constant $c_4 > 0$. In total, the Steklov–Rayleigh quotient of u_f on S_f is bounded as follows:

$$\begin{aligned}
 \mathcal{R}(u_f) &\leq \frac{1}{c_4 \epsilon^{n-1}} \left(c_1 \frac{\epsilon^{n-2}}{\sqrt{1 + \partial_2^2 \omega^2}} + c_2 \frac{\epsilon^{n-3} \ln(\partial_2 \omega + \sqrt{1 + \partial_2^2 \omega^2})}{\omega} + c_3 \partial_1 \epsilon^{n-3} \right) \\
 &= \frac{1}{c_4} \left(c_1 \frac{\epsilon^{-1}}{\sqrt{1 + \partial_2^2 \omega^2}} + c_2 \frac{\epsilon^{-2} \ln(\partial_2 \omega + \sqrt{1 + \partial_2^2 \omega^2})}{\omega} + c_3 \partial_1 \epsilon^{-2} \right).
 \end{aligned}$$

We are now ready to define the constants more precisely. By using the following:

$$\partial_1 = \epsilon^3, \quad \partial_2 = \epsilon^{3/2}, \quad \omega = \epsilon^{-3}, \quad \rho = \epsilon, \tag{4.6}$$

we obtain

$$\mathcal{R}(u_f) = \mathcal{O}(\epsilon^{1/2}) \quad \text{as } \epsilon \rightarrow 0.$$

We have proved that a local perturbation of U allows the construction of a local trial function with arbitrarily small Steklov–Rayleigh quotient. The proof of our main result is now an easy consequence.

Demostración del Teorema 1.1. Without loss of generality, we work under the assumption that $k \in \mathbb{N}$ is fixed. The general case will then follow by a standard diagonal selection argument. Let $k, j \in \mathbb{N}$ with j sufficiently large. Let $p_1, \dots, p_{k+1} \in B(p, \frac{1}{j}) \cap \Sigma$ be distinct points, and let ψ be the quasi-isometric chart from Lema 2.2. For each p_i , we follow the above construction to obtain deformation functions f_i that are disjointly supported, by taking $\epsilon > 0$ small enough, and trial functions u_i that have disjoint supports contained in $\psi(B(p, \frac{1}{j}) \cap M)$. By possibly choosing a smaller ϵ in the previous construction, we guarantee that the Rayleigh quotient of each u_i is smaller than $1/j$. Consider the deformation function $f = f_1 + \dots + f_{k+1}$ supported in $B(p, 1/j)$ and the perturbed manifold $M_j = \tilde{M}_f$. Taking the pullback

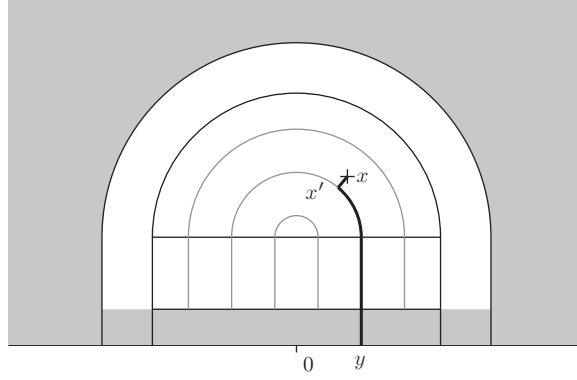


Figura 3. The set $\{x \mid f(x) = 0\}$ is shown in grey. The path in bold from an arbitrary $x \in \Omega$ to a point on $\partial\tilde{M}_f$ has length going to 0 as $\epsilon \rightarrow 0$.

by ψ , we obtain $k + 1$ trial functions $\psi^*(u_i)$ with disjoint supports and from Lema 2.1, their Rayleigh quotient is less than c/j where c is a constant depending on ψ . By the variational characterization (2.1) of the eigenvalue σ_k , we conclude that $\sigma_k(M_j) \leq c/j$.

Note that

$$\text{Vol}(S_f) = \int_{\Omega \cup D} \sqrt{1 + |\nabla f|^2} dx_1 \cdots dx_n \geq \text{Vol}(\Omega \cup D).$$

It follows from (4.2) and (4.4) that, on Ω , the following holds:

$$|\nabla f|^2 \leq 1 + \partial_2^2 \omega^2 = \epsilon^{-3}.$$

Similarly, it follows from (3.2) that on D ,

$$|\nabla f|^2 \leq c_5 \left(\frac{\partial_2^2}{\epsilon^2} + \partial_2^2 \omega^2 \right) = c_5 (\epsilon + \epsilon^{-3}),$$

where c_5 is a positive constant. It follows that

$$\begin{aligned} |\text{Vol}(S_f) - \text{Vol}(\Omega \cup D)| &= \int_{\Omega \cup D} \sqrt{1 + |\nabla f|^2} dx_1 \cdots dx_n - \text{Vol}(\Omega \cup D) \\ &= \mathcal{O}(\epsilon^{n-3/2}), \end{aligned}$$

which goes to 0 for $n \geq 2$. Finally, it is clear that the curvatures of $\partial\tilde{M}_f$ do not change, as M is kept fixed on some neighborhood of the boundary due to the localisation of f by the function γ , which vanishes near the boundary; see Figure 3 and the definition (3.2) of f . $\square$

Tabla 1. Este es el título de la tabla..

Projectile	Element 1			Element 2*		
	Energy	σ_{calc}	σ_{expt}	Energy	σ_{calc}	σ_{expt}
Element 3	990 A	1168	1547 ± 12	780 A	1166	1239 ± 100
Element 4	500 A	961	922 ± 10	900 A	1268	1092 ± 40

* Este es un ejemplo de nota al pie de tabla.

5. Conclusiones

Algunas conclusiones aquí.

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Conflictos de intereses. A statement about any financial, professional, contractual or personal relationships or situations that could be perceived to impact the presentation of the work — or ‘None’ if none exist

Declaración de disponibilidad de datos. A statement about how to access data, code and other materials allowing users to understand, verify and replicate findings — e.g. Replication data and code can be found in Harvard Dataverse: <https://doi.org/link>.

Estándares éticos. The research meets all ethical guidelines, including adherence to the legal requirements of the study country.

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